

Adam Ligocki

SOFTWARE ENGINEER · ROBOTICS · COMPUTER VISION · MACHINE LEARNING

I build the software and the machine learning that run inside robots and vehicles, and make them reliable enough for real industrial use. Thirteen years of software development in fields like autonomous driving, robotics and medical video. PhD in cybernetics and artificial intelligence. On my last engagement, as CTO, I took the product from an early prototype to a fleet of two hundred machines running unattended worldwide.

Open to permanent roles, contract and freelance work.

CONTACT

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CORE SKILLS

AI and machine learning

Deep learning for object detection, Multi-modal and thermal image learning, Dataset design and collection
Also: PyTorch, Model deployment with ONNX, TensorRT and CUDA, Transfer learning, Synthetic data and YOLO training pipelines, Computer vision for agriculture

Hardware

Sensor selection, mounting and calibration, Multi-sensor rig integration and time synchronisation, Embedded firmware running on production machines
Also: PCB and electronics design in KiCad, Robot platform bring-up, electronics to enclosure

Leadership

Technical leadership as CTO, Scaling a robot fleet from prototype to production, Rescuing an unreliable system
Also: Running a company

Programming

C++, Rust, Software architecture for embedded systems, CI/CD and release automation, MLOps and model deployment to production, Simulation and replay tooling
Also: Python, Docker and containerised builds, AI-assisted development with Claude Code and Copilot

Robotics

Sensor fusion, SLAM and 3D mapping, LiDAR and point cloud processing, ROS and ROS 2, Autonomous driving
Also: GNSS, RTK and IMU integration, UAV autopilot and control, Agricultural robotics

LANGUAGES

Czech (native)
Polish (native)
English (C1, fluent)
Ukrainian (B2)

EXPERIENCE

AI specialist, then CTO

ULLMANNA s.r.o.

Rebuilt the weeding control system and led the technical team.

- Real-time vision-guided control system on embedded hardware
- From prototype to hundreds of machines worldwide

Rust · C++ · Python · Computer vision · Machine learning · System architecture

2023 to 2026

C++ developer, sensor fusion and computer vision

ARTIN Group (RoboAuto)

Sensor fusion, computer vision and C++ for autonomous vehicles.

- Sensor fusion and computer vision for the first Czech self-driving car, built on affordable sensors
- Perception work on the first Czech autonomous train, delivered with AŽD Praha
- Both systems were publicly demonstrated and covered in national media

C++ · Computer vision · Sensor fusion · LiDAR · Autonomous driving

2017 to 2023

Researcher

Brno University of Technology

Research in sensor fusion and robotic environment mapping.

- Brno Urban Dataset: over 350 km of multi-sensor recordings, published at IEEE ICRA 2020
- Atlas Fusion: real-time multi-sensor fusion framework, later rebuilt on ROS 2

C++ · ROS · ROS 2 · Sensor fusion · LiDAR · Thermal imaging · Deep learning · CNN

2017 to 2026

Lecturer and external pedagogue

Brno University of Technology

- Ten courses in programming, robotics and computer vision; author of a hands-on robotics course

C++ · Python · ROS · Robotics education

2018 to 2026

Software developer

Gatema Medical

Operating-theatre video streaming, on macOS and iOS.

- Live 4K video capture, switching and streaming for operating theatres
- The product line continues today as the Artemis operating-room system

C++ · macOS · iOS · Video streaming · Real-time video

2013 to 2017

EDUCATION

2017 to 2022 · Brno University of Technology

Ph.D. in Cybernetics and Artificial Intelligence. Thesis on sensor fusion for 3D dynamic mapping.

2018 · Tampere University of Technology

Erasmus research exchange, Finland. First international research experience.

PUBLICATIONS

18 peer-reviewed publications, 2013 to 2024, including IEEE ICRA 2020 and two papers in Sensors.

Brno Urban Dataset - The New Data for Self-Driving Agents and Mapping Tasks

IEEE International Conference on Robotics and Automation (ICRA), 2020

Fully Automated DCNN-Based Thermal Images Annotation Using Neural Network Pretrained on RGB Data

Sensors, 2021

Atlas Fusion - Modern Framework for Autonomous Agent Sensor Data Fusion

14th International Conference ELEKTRO, IEEE, 2022

Robotic Template Library

Journal of Open Research Software, 2021